Heavy-Quark Spin Symmetry Violation Effects in Charmed Baryon Production

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Introduction

Background

Why do we study Charmed Baryon Production ?

Containing a charm quark in heavy hadron exhibits unique properties, providing a better understanding of Quantum Chromodynamics (QCD).

- 1974: Discovery of J/Ψ marked the beginning of research into the production of charmonium ($c\bar{c}$) state.
  - Labs: Tevatron, HERA, BABAR, Belle, BES.
- 1975: First charmed baryon, Λ_c^+, discovered in neutrino interactions.
  - Labs: FOCUS, SELEX, ARGUS, CLEO, BABAR, BELLE.
- In the near future, the laboratories aim for highly accurate spectroscopy of charmed hadrons.
  - Japan Proton Accelerator Research Complex (J-PARC), Japan
  - AntiProton Annihilation at Darmstadt ($\bar{P}ANDA$) at FAIR, Germany

Heavy-Quark Spin Symmetry (HQSS)

- HQSS is an approximate symmetry of QCD in the limit m_b,$m_c \gg \Lambda_{QCD}$,
  - Light quarks do not couple to the spin of the heavy quark.

$$m_{D^*} - m_D \approx \frac{1}{m_c},\ D(J^P = 0^+),\ D^*(J^P = 1^+)$$
$$m_{B_c^*} - m_{B\,c} \approx \frac{1}{m_c},\ B(J^P = 1/2^+),\ D^*(J^P = 3/2^+)$$

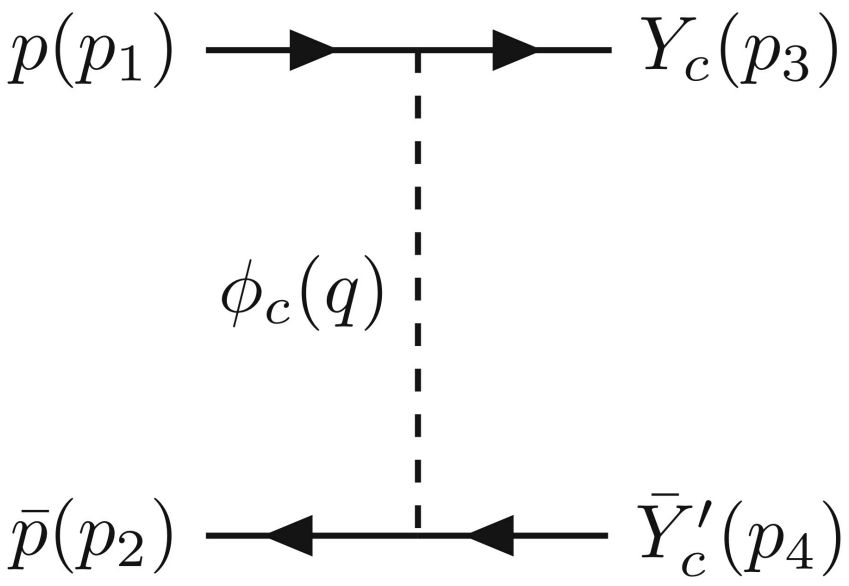
- HQSS implies that pseudoscalar (D) and vector D mesons (D^*), as well as charm baryon sextets with spins 1/2 (B_c) and 3/2 (B_c^*), form degenerate states.
- Degeneracy depends on the spin-flip of the charm quark.

$$\begin{pmatrix} D^* \\ D \end{pmatrix}, \quad \begin{pmatrix} B^* \\ B \end{pmatrix}$$

Objectives

To calculate the Conserving & Violating contributions of the HQSS of the charmed baryons production Y_c, $\bar{Y}'_c$ processes with the exchanged particles $\phi_c = D, D^*$ mesons.

$$p\bar{p} \to \Lambda_c\bar{\Lambda}_c, \Sigma_c\bar{\Sigma}_c, \Sigma_c^*\bar{\Sigma}_c^*, \Sigma_c^*\bar{\Sigma}_c, \Sigma_c\bar{\Lambda}_c, \Sigma_c^*\bar{\Lambda}_c.$$



Methodology

Methodology

1. Construct the $SU(2)_f$ effective Lagrangains
2. Construct CHQSS & VHQSS Lagrangians
3. Matching relation between $SU(2)_f$ & CHQSS + VHQSS, by relating their coupling constants
4. Calculate the scattering amplitudes& differential cross-sections for $p\bar{p} \rightarrow \Lambda_c\bar{\Lambda}_c, \Sigma_c\bar{\Sigma}_c, \Sigma_c^*\bar{\Sigma}_c^* \Sigma_c^*\bar{\Sigma}_c, \Sigma_c\bar{\Lambda}_c, \Sigma_c^*\bar{\Lambda}_c,$
5. Evaluate & compare the contribution of conserving & violating parts

The $SU(2)_f$ Effective Lagrangian

The basic building blocks:

Baryons

$$N_a = \begin{pmatrix} p \\ n \end{pmatrix}_a, J^P = \frac{1}{2}^+; \quad \Lambda_{(c)} = \Lambda^+_{(c)}, \quad J^P = \frac{1}{2}^+$$

$$\Sigma_{(c)ab} = \begin{pmatrix} \Sigma^{++}_{(c)} \\ \Sigma^{+}_{(c)} \\ \Sigma^{0}_{(c)} \end{pmatrix}_{ab}, J^P = \frac{1}{2}^+; \quad \Sigma^{\mu}_{(c)ab} = \begin{pmatrix} \Sigma^{++}_{(c)\mu} \\ \Sigma^{+}_{(c)\mu} \\ \Sigma^{0}_{(c)\mu} \end{pmatrix}_{ab}, J^P = \frac{3}{2}^+$$

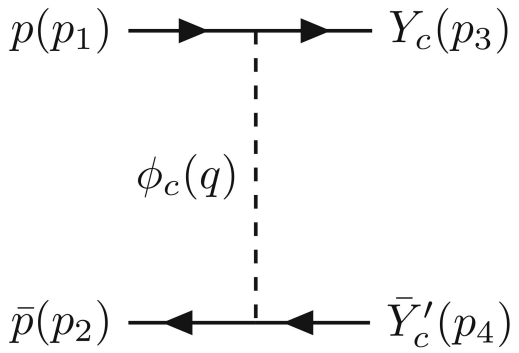
Mesons

$$D^i_a = \begin{pmatrix} D^0 & D^+ \end{pmatrix}_a, J^P = 0^+; \quad D^i_{\mu a} = \begin{pmatrix} D^0_\mu & D^+_\mu \end{pmatrix}_a, J^P = 1^+$$

The $SU(2)_f$ Effective Lagrangian

At chiral power counting of Q^0 and Q^1.

$$
\begin{aligned}
\mathcal{L} &= \mathcal{L}^P + \mathcal{L}^A + \mathcal{L}^V + \mathcal{L}^T \\
\mathcal{L}^P &= \mathcal{L}_{DN\Sigma_c} + \mathcal{L}_{DN\Lambda_c} + \text{h.c.}\,, \\
\mathcal{L}^A &= \mathcal{L}_{DN\Sigma_c} + \mathcal{L}_{DN\Lambda_c} + \mathcal{L}_{DN\Sigma_c^*} + \text{h.c.}\,, \\
\mathcal{L}^V &= \mathcal{L}_{D^*N\Sigma_c} + \mathcal{L}_{D^*N\Lambda_c} + \mathcal{L}_{D^*N\Sigma_c^*} + \mathcal{L}_{D^*N\Sigma_c^*} + \text{h.c.}\,, \\
\mathcal{L}^T &= \mathcal{L}_{D^*N\Sigma_c} + \mathcal{L}_{D^*N\Lambda_c} + \mathcal{L}_{D^*N\Sigma_c^*} + \text{h.c.},
\end{aligned}
$$



$$p\bar{p} \to \Lambda_c\bar{\Lambda}_c, \Sigma_c\bar{\Sigma}_c, \Sigma_c^*\bar{\Sigma}_c^*, \Sigma_c^*\bar{\Sigma}_c, \Sigma_c\bar{\Lambda}_c, \Sigma_c^*\bar{\Lambda}_c$$

CHQSS & VHQSS Lagrangains

Building blocks of conserving & violating parts.

C:	$\Gamma^\mu = 1, \gamma_5,$
V:	$\Gamma^\mu = \gamma^\mu, \gamma^\mu\gamma_5, \sigma^{\mu\nu}$

Taking into account for all of the super-multiplet fields of the HQSS.

$$\mathcal{L}_{\text{CHQSS}} = \frac{1}{2} c_1 \left\langle \bar{T}^\mu_{ab} \gamma_5 N_b H_a \gamma_\mu + \text{h.c.} \right\rangle + \frac{1}{2} c_2 \left\langle \bar{T} \gamma_5 N_a H_a \gamma_5 + \text{h.c.} \right\rangle \tag{1}$$

$$\begin{aligned} \mathcal{L}_{\text{VHQSS}} = & -\frac{1}{4} b_1 \left\langle \bar{T}^\mu_{ab} \gamma^\nu N_b H_a \gamma_\mu \gamma_\nu \gamma_5 + \text{h.c.} \right\rangle + \frac{1}{2} b_2 \left\langle \bar{T} \gamma^\mu N_a H_a \gamma_\mu + \text{h.c.} \right\rangle \\ & + \frac{1}{4} b_3 \left\langle \bar{T} \gamma^\mu \gamma^\nu \gamma_5 N_a H_a \gamma_\mu \gamma_\nu \gamma_5 + \text{h.c.} \right\rangle . \end{aligned} \tag{2}$$

CHQSS & VHQSS Lagrangains

Then, after tracing all fields, it read

$$\mathcal{L}_{\text{CHQSS}} = \frac{1}{\sqrt{3}} c_1 \bar{\Sigma}_{(c,+)ab} \gamma^\mu N_b D_{\mu a,+} + c_1 \bar{\Sigma}^\mu_{(c,+)ab} \gamma_5 N_b D_{\mu a,+} + c_2 \bar{\Lambda}^+_{(c,+)} i\gamma_5 N_a D_{a,+} + \text{h.c.}\,, \tag{3}$$

$$\begin{aligned}\mathcal{L}_{\text{VHQSS}} = & \frac{\sqrt{3}}{2} b_1 \bar{\Sigma}_{(c,+)ab} i\gamma_5 N_b D_{a,+} + \frac{i}{\sqrt{3}} b_1 \bar{\Sigma}_{(c,+)ab} \sigma^{\mu\nu} N_b D_{\mu a,+} v_\nu \\ & - \frac{i}{2} b_1 \epsilon_{\mu\nu\alpha\beta} \bar{\Sigma}^\mu_{(c,+)ab} \gamma^\nu N_b D^\alpha_{a,+} v^\beta + b_2 \bar{\Lambda}^+_{(c,+)} \gamma^\mu N_a D_{\mu a,+} \\ & + 2 b_3 \bar{\Lambda}^+_{(c,+)} i\gamma_5 N_a D_{a,+} + i b_3 \bar{\Lambda}^+_{(c,+)} \sigma^{\mu\nu} N_a D_{\mu a,+} v_\nu + \text{h.c.}.\end{aligned} \tag{4}$$

Matching relation

intermediate particle		$SU(2)_f$			CHQSS			VHQSS		
		$\Lambda_{(c)}$	$\Sigma_{(c)ab}$	$\Sigma^{\mu}_{(c)ab}$	$\Lambda^{+}_{(c,+)}$	$\Sigma_{(c,+)ab}$	$\Sigma^{\mu}_{(c,+)ab}$	$\Lambda^{+}_{(c,+)}$	$\Sigma_{(c,+)ab}$	$\Sigma^{\mu}_{(c,+)ab}$
D_a	pseudo-scalar (P)	g_2	g_1		c_2			b_3	b_1	
	axial-vector (A)	g_4	g_3	g_5						
$D_{\mu a}$	vector (V)	f_2	f_1	f_3		c_1	c_1	b_2		
$D_{\mu\nu a}$	tensor (T)	h_2	h_1	h_3				b_3	b_1	b_1

Table: The coupling constants of the $SU(2)_f$, *conserving*-HQSS and *violating*-HQSS Lagrangains.

CHQSS

$$c_1 = \frac{\sqrt{3}}{2} f_1, \quad c_1 = \frac{1}{2} f_3, \quad f_3 = \sqrt{3} f_1, \quad c_1 = \frac{1}{2} h_3, \quad c_2 = g_2 + g_4,$$

VHQSS

$$b_1 = \frac{g_1 + g_3}{2\sqrt{3}}, \quad b_1 = -\frac{\sqrt{3}}{8} h_1, \quad b_1 = \frac{1}{8} f_4, b_2 = f_2, \quad b_3 = \frac{g_2 + g_4}{8}, \quad b_3 = \frac{1}{8} h_2$$

Results &Discussion

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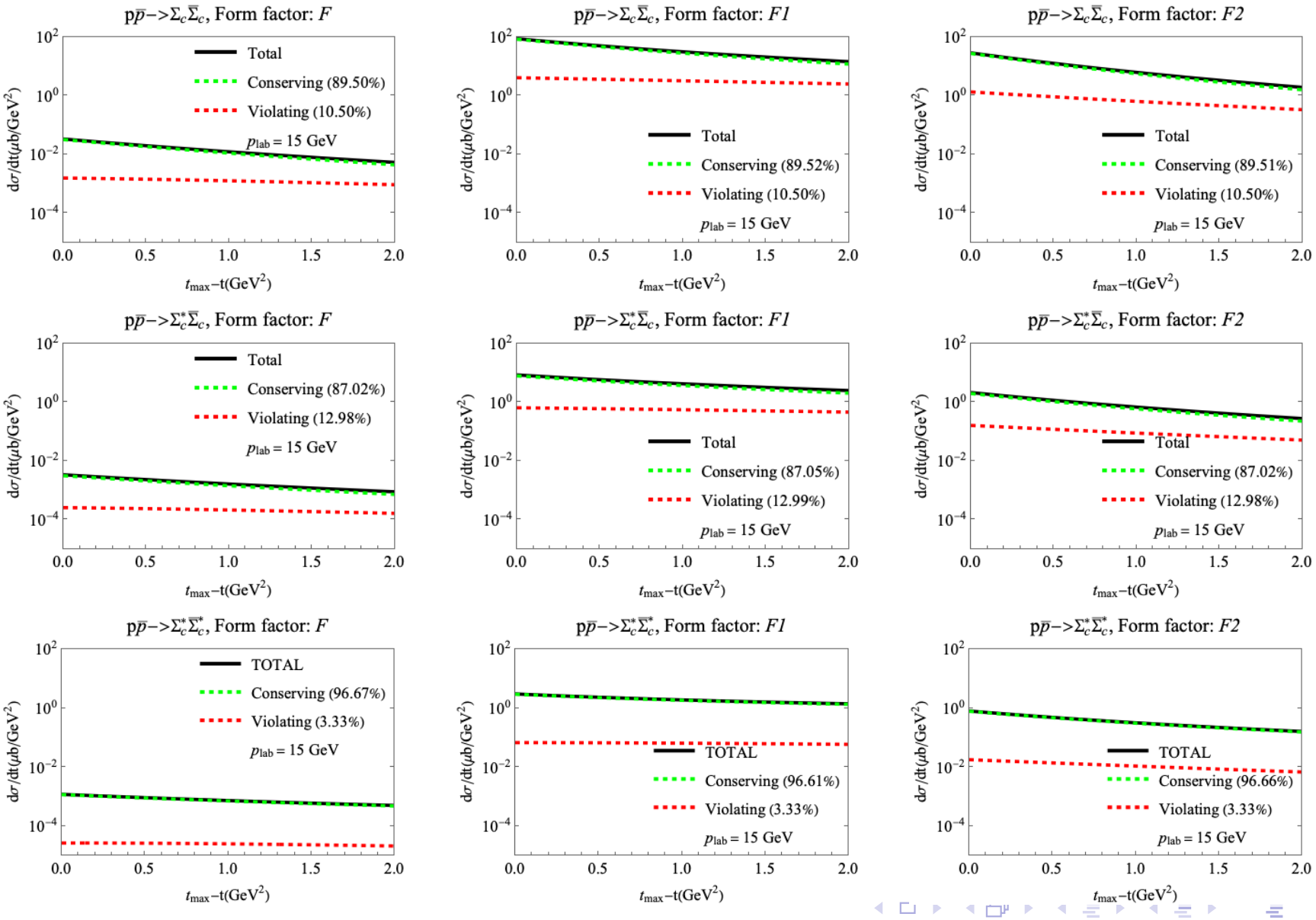


Conclusion

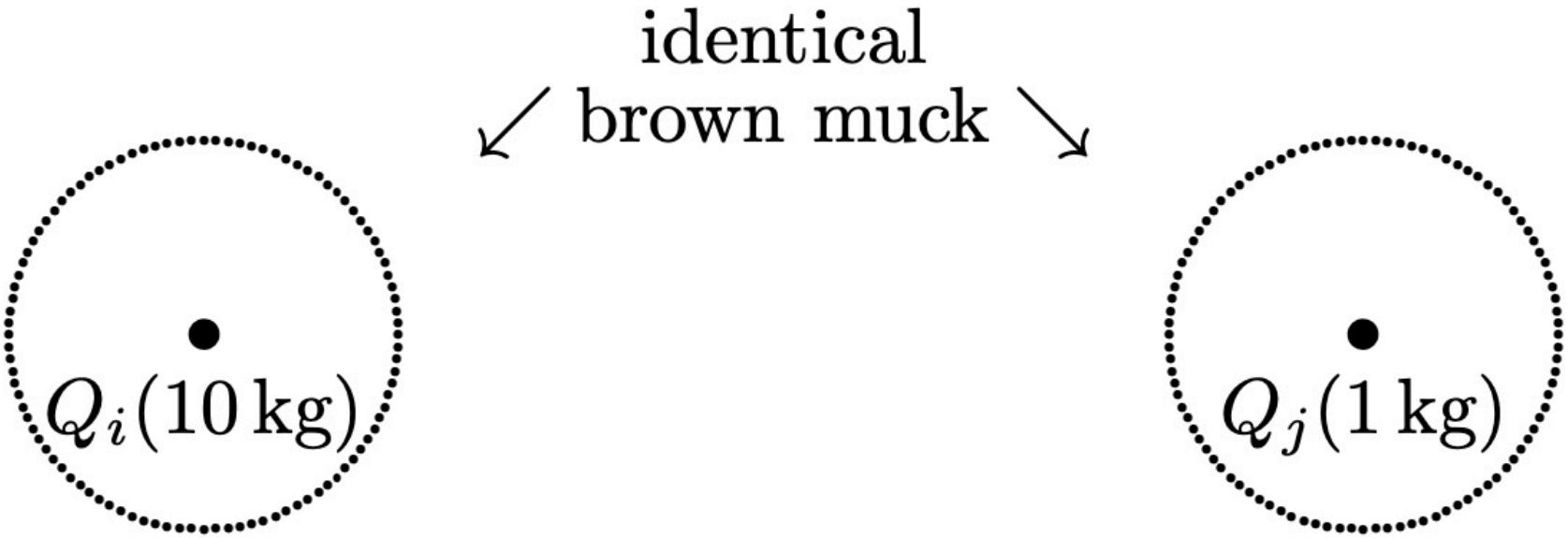
Conclusion

- Introduction of charmed baryon productions and construct *conserving* & *violating* HQSS effective L.
- Calculation the differential cross-sections for $p\bar{p} \to Y_c \bar{Y}_c'$.
- The differential cross sections of Σ_c and Σ_c^* productions, are mainly contributed by conserving HQSS. The order of magnitudes lies between 10^{-4} and 10^2
- The processes of Λ_c productions is ongoing.

Thank you for your kind attention.

Backup slides

Heavy quark Q_i have a mass of 10kg and quark Q_j have a mass of 1kg.



- HQSS is an approximate symmetry of QCD in the limit $m_b, m_c \gg \Lambda_{QCD}$,
- Light quarks do not couple to the spin of the heavy quark.

$$m_{D^*} - m_D \approx \frac{1}{m_c},\ D(J^P = 0^+),\ D^*(J^P = 1^+)$$

$$m_{B_c^*} - m_{B_c} \approx \frac{1}{m_c},\ B(J^P = 1/2^+),\ D^*(J^P = 3/2^+)$$

Backup slides

Λ_c^+mass $\approx 2284.9 MeV$
D mass $\approx 1869 MeV$
D^* mass $\approx 2006 MeV$
c quark $\approx 1275 MeV$
Σ_c^+

$\Sigma_c(2455)^{++}$ MASS

VALUE (MeV)
2453.97±0.14 OUR FIT

$\Sigma_c(2455)^{+}$ MASS

VALUE (MeV)
$2452.65^{+0.22}_{-0.16}$ OUR FIT

$\Sigma_c(2455)^{0}$ MASS

VALUE (MeV)
2453.75±0.14 OUR FIT

$\Sigma_c(2520)^{++}$ MASS

VALUE (MeV)	*EVT*
2518.41±0.22 OUR FIT	
• • • We do not use the fo	
2530 ±5 ±5	

[1] AMMOSOV 93 sees a c

$\Sigma_c(2520)^{+}$ MASS

VALUE (MeV)
$2517.4^{+0.7}_{-0.5}$ OUR FIT

$\Sigma_c(2520)^{0}$ MASS

VALUE (MeV)
2518.48±0.21 OUR FIT

Backup slides

3-point vertices effective chiral Lagrangian
There are 15 terms at the chiral power counting order Q^0 and Q^1 including the Clifford algebra elements:

$$\Gamma = \gamma^\mu, \gamma^\mu\gamma_5, \sigma^{\mu\nu}, 1 \text{ for the chiral power counting order } Q^0;$$

and

$$\Gamma = \gamma_5 \quad \text{for the chiral power counting order } Q^1.$$

HQSS Conservation & Violation

$$\mathcal{L} = \bar{H}^{\mu}\,\Gamma_{\mu} H^{\nu}$$

C:	$\Gamma^{\mu} = 1, \gamma_5,$
V:	$\Gamma^{\mu} = \gamma^{\mu}, \gamma^{\mu}\gamma_5, \sigma^{\mu\nu}$

As a consequent, the pseudoscalar and vector D mesons as well as spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ baryons form degenerate states, which can be defined by

$$\begin{aligned}
H_a &= \left(\frac{1+\not{v}}{2}\right)\left(D_{\mu a,+}\gamma^{\mu} + i\gamma_5 D_{a,+}\right) &&, \bar{H}_a = \gamma_0 H_a^{\dagger}\gamma_0, \\
T^{\mu}_{ab} &= \frac{1}{\sqrt{3}}\left(\gamma^{\mu} + v^{\mu}\right) i\gamma_5 \left(\frac{1+\not{v}}{2}\right)\Sigma_{(c,+)ab} + \left(\frac{1+\not{v}}{2}\right)\Sigma^{\mu}_{(c,+)ab} &&, \bar{T}^{\mu}_{ab} = \left(T^{\mu}_{ab}\right)^{\dagger}\gamma_0 \\
T &= \frac{1+\not{v}}{2}\Lambda_{(c,+)} &&, \bar{T} = T^{\dagger}\gamma_0.
\end{aligned} \tag{5}$$

These super-multiplet heavy-quark hadronic fields, as a building blocks for the

Backup slides

In the large heavy-quark mass limit, single heavy-quark hadrons can be considered as slowly varying fields, $D_\pm(x)$, $D^\mu_\pm(x)$, $\Sigma_{(c,\pm)}(x)$, $\Sigma^\mu_{(c,\pm)}(x)$ and $\Lambda_{(c,\pm)}$ (we drop the flavor indices here for simplicity). Then, D meson and charmed baryons fields are decomposed as the following expressions,

$$\begin{aligned}
D(x) &= e^{-i\,(v\cdot x)\,M_c}\,D_+(x) + e^{+i\,(v\cdot x)\,M_c}\,D_-(x)\,,\\
D^\mu(x) &= e^{-i\,(v\cdot x)\,M_c}\,D^\mu_+(x) + e^{+i\,(v\cdot x)\,M_c}\,D^\mu_-(x)\,,\\
\Sigma_c(x) &= e^{-i\,(v\cdot x)\,M_{\Sigma_c}}\,\Sigma_{c,+}(x) + e^{+i\,(v\cdot x)\,M_{\Sigma_c}}\,\Sigma_{c,-}(x)\,,\\
\Sigma^\mu_c(x) &= e^{-i\,(v\cdot x)\,M_{\Sigma^*_c}}\,\Sigma^\mu_{c,+}(x) + e^{+i\,(v\cdot x)\,M_{\Sigma^*_c}}\,\Sigma^\mu_{c,-}(x)\,,\\
\Lambda_c(x) &= e^{-i\,(v\cdot x)\,M_{\Lambda_c}}\,\Lambda_{c,+}(x) + e^{+i\,(v\cdot x)\,M_{\Lambda_c}}\,\Lambda_{c,-}(x)\,,
\end{aligned} \tag{9}$$

with a 4-velocity v normalized by $v^2 = 1$. The mass parameters M_c, M_{Σ_c}, $M_{\Sigma^*_c}$, and M_{Λ_c} are the charm quark, Σ_c spin-1/2, Σ_c spin-3/2, and Λ_c masses respectively.